Remote Controller with Fail-Safe Update

Concept of Operations

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1.0 Introduction

1.1 Project Description

1.1.1 Background

Purpose: The System lets an operator on the ground command, monitor, and update a controller in space that nobody can reach, and keeps that controller working between contacts.

Need: A controller that nobody can reach must stay operable, observable, and correctable for its whole life. Operation, observability, and maintenance including in-flight modification over the specified lifetime are three areas a space software specification must cover [3, Annex B, items 5.11–5.13, p. 113].

Worst-Case Failure: A failed update can leave the controller unable to run any firmware and unable to accept new firmware. Without physical access, that state is permanent.

Rationale: The loss has happened. NASA records a mission terminated after an in-flight software upload omitted a patch for a defective countdown timer; its recommendations include test-bed verification of in-flight software updates [4, Lesson 0559, p. 199].

1.1.2 Assumptions and Constraints

C-1 The ground cannot talk to the controller directly. Every command, report, and firmware image passes through the onboard computer.

C-2 All software is built to NASA's software engineering requirements. Rigor per part is set by its software class [5, Sec. 3.5.1, p. 24; App. C].

C-3 One engineer builds and checks the System. No independent team reviews the work.

C-4 The System runs on commercial development boards, not on spacecraft hardware. It can show how the System behaves, not that it would survive launch or space.

A-1 Once launched, physical access to the onboard computer or controller is impossible; all fixes and updates must arrive by radio.

A-2 The controller can lose power at any moment, even halfway through writing new firmware. The System must survive that without losing the controller [2, App. A, Table 2, p. 63].

A-3 The ground can reach the spacecraft only during line-of-sight contact windows **[TBD-5]**. Between contacts the System must work on its own.

A-4 One person operates the ground station. No second person checks a command before it is sent, so the System itself must refuse wrong commands.

A-5 Radiation in space can flip single bits in memory, so stored firmware can change on its own. The System must not run firmware that has changed [2, App. A, Table 2, p. 63].

Note. *A constraint (C) is a limit the System must work within; it is imposed, not chosen. An assumption (A) is a condition taken as true without proof; the design depends on it, and if it proves false the design may be wrong.* ***[TBR-**n**]*** *marks a proposal the author has not yet accepted;* ***[TBD-**n**]*** *marks a value not yet known. Both are listed in App. C.*

1.2 Overview of the Envisioned System

1.2.1 Overview

The System comprises two spatially separated operational segments. The **ground segment** on Earth includes the operator, the mission operations center running ground software, and the ground station antenna [6, Sec. 11.2, p. 305]. The **space segment** in orbit consists of the onboard computer and the controller, which operate without physical access.

Communication between segments is strictly established via a radio link during line-of-sight contact windows [**TBD-5**]. Outside contact windows, the space segment operates autonomously. Within the spacecraft, the onboard computer communicates with the controller over the data link during normal operation. If the controller's main firmware fails to execute, the onboard computer utilizes the reprogramming link to restore functional firmware independently of the controller's primary software. Figure 1 illustrates this segment layout.

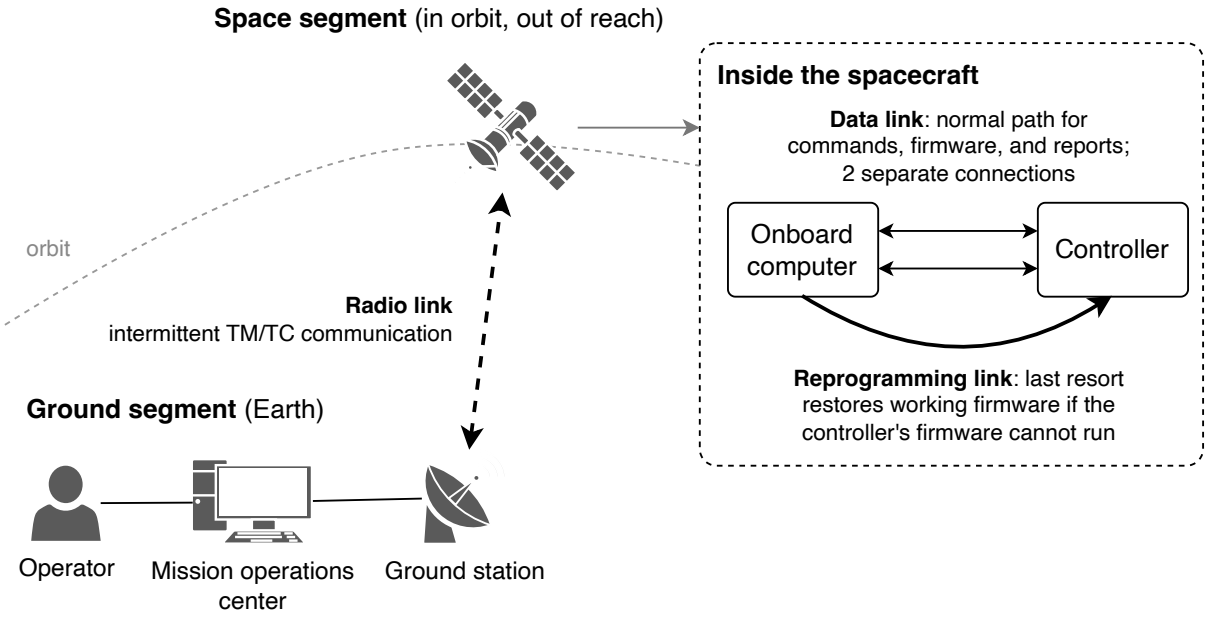


Figure 1: Ground and space segments, joined only by the radio link during contact windows. Inside the spacecraft, the data link handles normal communication, while the reprogramming link restores working firmware if the controller's firmware cannot run.

1.2.2 System Scope

The system context defines the boundary of the System: the system of interest, its users, and the external elements that interface with it [7, Sec. 8.3, p. 24].

Context Element	System Boundary
System of Interest	Onboard computer, controller, ground software in the MOC
Users	Ground operator
External Elements	Radio equipment, spacecraft power supply

- **Enabling systems:** firmware build tools and bench test setups required for development and verification, external to operational scope.
- **In scope:** commanding, telemetry, firmware updates, and autonomous fault handling between contact windows [**TBR-5**].
- **Out of scope:** onboard computer software updates [**TBR-7**], and the radio equipment, which is used as given.

2.0 Documents

2.1 Applicable Documents

[1] NASA, *NASA Systems Engineering Handbook*, NASA/SP-2016-6105 Rev 2, 2016. This outline follows its App. S, pp. 251–253, tailored per its Sec. 3.11.1, p. 34: Section 7.0 not applicable; App. C added for open items per its App. C, pp. 197–198.

2.2 Reference Documents

[2] NASA, *Software Assurance and Software Safety Standard*, NASA-STD-8739.8B, 2022. Off-nominal scenarios cite App. A, Table 2.
[3] ESA, *Space engineering — Software*, ECSS-E-ST-40C Rev. 1, 30 April 2025.
[4] NASA, *Fault Management Handbook*, NASA-HDBK-1002, Draft 2, 2 April 2012. A draft, not a released standard.
[5] NASA, *NASA Software Engineering Requirements*, NPR 7150.2D.
[6] NASA, *State-of-the-Art of Small Spacecraft Technology*, Chapter 11, "Ground Data Systems and Mission Operations", 2026 edition. Ground system architecture in Sec. 11.2 and Fig. 11.2, pp. 305–306.
[7] NASA, *NASA Systems Modeling Handbook for Systems Engineering*, NASA-HDBK-1009A, 12 March 2025.

3.0 Description of Envisioned System

3.1 Needs, Goals and Objectives

NEED A controller that nobody can reach must stay operable, observable and correctable for its whole life.

G-1 **Operability:** the controller keeps running and acts on the operator's commands.
- **O-1** Every valid operator command is executed. Valid means from the operator, allowed in the current state, with possible values, and not expired.
- **O-2** Firmware and commands not from the operator are never acted on.
- **O-3** With one data-link connection lost, commands, telemetry and updates continue.
- **O-4** No fault found between contacts goes unreported [**TBR-5**].

G-2 **Observability:** the ground can know what the System is doing, how healthy it is, and what happened between contacts.
- **O-5** No report is lost without the ground being told.

G-3 **Correctability:** the controller's firmware can be replaced for its whole life without losing the controller.
- **O-6** No update, even one interrupted by a single fault, leaves the controller without runnable firmware.

3.2 Overview of System and Key Elements

Operator The user. Decides and sends commands, releases new firmware, and reads the reports.

Ground software In the mission operations center. Turns the operator's decisions into commands and firmware ready to send, checks them before they leave, and keeps every report received.

Onboard computer Checks commands and firmware and passes them to the controller, stores the controller's reports between contacts, watches the controller, and restores it when its firmware cannot run.

Controller Runs its firmware, acts on commands, reports its state, and returns to working firmware on its own when new firmware fails.

The ground station's radio equipment and the spacecraft power supply are outside the System; it uses them as given.

3.3 Interfaces

External interfaces

Operator interface Human. Ground software to operator. Allows the flight operator to issue telecommands, authorize firmware deployments, and monitor system telemetry.

Radio link Radio frequency. Ground software to onboard computer via external radio equipment. Provides intermittent telemetry and telecommand (TM/TC) transfer exclusively during contact windows.

Power Electrical. Spacecraft power subsystem to onboard computer and controller. Subject to unannounced power interruptions at any time (assumption A-2).

Interfaces between elements

Data link Digital data. Onboard computer to controller. Nominal interface for commands, telemetry, and firmware updates while controller firmware is running. Uses two separate connections, so a single connection failure does not interrupt operations.

Reprogramming link Digital data. Onboard computer to controller. Recovery interface used as a last resort to restore functional firmware independently of the controller's main software.

3.4 Modes of Operations

Life-cycle modes of the System

Test Ground bench execution with simulated radio link degradation and intentional fault injection to demonstrate operational scenarios.

Operation Active mission execution using the controller modes detailed below.

Disposal Not applicable; limited to bench demonstration scope.

Controller modes during operation

OPERATE Nominal state executing confirmed firmware. New firmware images can be staged in the background during execution [**TBR-6**].

TRIAL Entered upon transitioning to updated firmware. Executes the new image for a bounded evaluation window [**TBD-6**]. If the new image is judged good, the update is committed and the controller returns to OPERATE; if not, it enters FALLBACK.

FALLBACK Entered following a failed firmware trial or boot failure. Reverts execution to the previous known-good firmware image or the last-resort firmware.

RECOVERY Entered when no operational firmware image can be executed. Controller software execution is halted while the onboard computer restores functional firmware via the reprogramming link.

3.5 Proposed Capabilities

Telemetry (serves G-2 Observability)

Capability-1 Generate periodic health reports detailing operational status, active firmware version, uptime, reset cause, and thermal telemetry.

Capability-2 Generate asynchronous event-driven reports immediately upon significant state changes or operational events.
Capability-3 Apply time tags to all generated telemetry reports.
Capability-4 Store telemetry reports during out-of-contact periods for downlink during subsequent contact windows, including notification of any unrecoverable telemetry data loss.
Capability-5 Monitor and report health status for each connection of the data link.

Command handling (serves G-1 Operability)
Capability-6 Execute telecommands originating exclusively from the authorized ground operator and reject replay attacks.
Capability-7 Acknowledge command reception, disposition (executed or rejected with cause), and completion status for every incoming telecommand.
Capability-8 Reject telecommands invalid for the current operational mode.
Capability-9 Run a repeated telecommand only once.
Capability-10 Enforce a two-step "arm and execute" protocol for critical operations.
Capability-11 Reject telecommands containing out-of-bounds parameter values.
Capability-12 Allow the operator to abort an in-progress firmware update while preserving current operational firmware.
Capability-13 Support time-tagged command execution and reject expired commands.
Capability-14 Maintain a command log recording all incoming telecommands and execution responses.

Fault handling (serves G-1 Operability) [**TBR-5**]
Capability-15 Detect controller unresponsiveness and initiate autonomous recovery sequences.
Capability-16 Automatically fail over to the redundant data-link connection upon single-path failure.
Capability-17 Restart crashed onboard processes and continue where they stopped.
Capability-18 Log all detected fault events and executed recovery actions for subsequent downlink.

Update (serves G-3 Correctability)
Capability-19 Support reliable firmware image reception across intermittent radio contact windows.
Capability-20 Validate firmware integrity, source authentication, target compatibility, and version policy (preventing unauthorized rollbacks).
Capability-21 Transfer validated firmware images to the controller.
Capability-22 Transition execution to newly uploaded firmware, with the activation trigger and scheduling criteria [**TBD-3**].
Capability-23 Evaluate new firmware operational validity post-boot; who judges is open [**TBD-4**].
Capability-24 Automatically roll back to operational firmware if trial evaluation fails, without ground intervention.
Capability-25 Downlink firmware update outcome reports to the ground segment.

Test (verification only)
Capability-26 Provide controlled, deliberate radio link degradation during bench testing.
Capability-27 Support command-driven injection of power interrupts and data-link disconnection faults [**TBR-10**].

Disposal: none; operational scope limited to bench demonstration.

4.0 Physical Environment

This section specifies the environment the System encounters across its life cycle, encompassing simulated mission operations and laboratory integration and testing. For each environmental condition, the expected system response is defined as nominal operation, degraded performance tolerance, or survivability [1, App. S, p. 252].

Phase	Environmental condition	Response	Shown on bench
Operation	Thermal range **[TBD-9]**	OPERATE	NO
Operation	Vacuum	OPERATE	NO
Operation	Radiation: single-event upset / memory bit errors	SURVIVE	SIMULATED
Operation	Electrical: power interruption at any instant	SURVIVE	YES
Launch	Mechanical: vibration and launch loads	OPERATE	NO
Integration and Test	Laboratory bench: ambient room temperature and pressure	OPERATE	YES

Legend: response

OPERATE	All functions continue normally.
TOLERATE	Partial functional degradation occurs, but no function is permanently lost.
SURVIVE	Operational functions temporarily halt and automatically recover upon removal of the condition.

Legend: shown on bench

YES	The condition itself is reproduced on the bench.
SIMULATED	The condition's effect is injected on purpose; the condition itself is not reproduced.
NO	Beyond what bench hardware can show (constraint C-4).

Transportation: not applicable; system transportation is outside the scope of bench demonstration.

5.0 Support Environment

Operational planning	The operator plans each contact in advance: which commands to send, which firmware to upload, and which reports to read. Contact times follow from the orbit **[TBD-5]**.
Commanding and uploads	The operator decides commands; the ground software prepares and checks them. They are sent during a contact or scheduled to run later. Firmware is built and released on the ground; one upload may span several contacts.
Maintenance and repair	None in orbit; there is no physical access (A-1). Faults are handled by the System itself, or by the operator through commands and new firmware.
Sparing	No hardware spares in orbit. The onboard computer keeps the last known-good firmware for the controller, and the controller keeps its last-resort firmware.
Upgrades	New controller firmware is the only upgrade path; update rate **[TBD-8]**. The onboard computer's own software is not upgraded **[TBR-7]**.

Continued support The engineer stays available to build and release new firmware for the life of the mission [**TBR-11**].

6.0 Operational Scenarios

Each scenario follows one thread of events from trigger to outcome [1, App. S, pp. 252–253]. Every scenario below is demonstrated on hardware and recorded. Sources in brackets are items of NASA's list of software causes of hazards [2, App. A, Table 2].

6.1 Nominal Conditions

Scenario-100 Firmware update.

1. The operator releases new firmware; the ground software checks it and prepares the upload.
2. Over one or more contacts, the firmware goes up the radio link to the onboard computer.
3. The onboard computer checks that it is undamaged, from the operator, built for this controller, and not older than the minimum version.
4. The onboard computer delivers it to the controller over the data link; the controller keeps working meanwhile [**TBR-6**].
5. The controller switches to the new firmware and enters TRIAL [**TBD-3**].
6. The new firmware is judged good within the trial time [**TBD-6**] [**TBD-4**]; the update is committed and the controller returns to OPERATE.
7. The outcome is stored and sent down at the next contact.

Scenario-110 Telemetry between contacts.

1. The controller sends regular health reports, and an event report as soon as something important happens.
2. The onboard computer time-tags and stores every report.
3. At the next contact, all stored reports go down the radio link.

Scenario-120 Command during a contact.

1. The operator decides a command; the ground software prepares and checks it.
2. During a contact, the command goes up the radio link.
3. The onboard computer checks that it is from the operator, new, allowed in the current mode, and has possible values. A dangerous command must first be armed by a separate command.
4. The command is executed. The answers (received, then done with the result) go down, and the command and its answers are logged.

Scenario-130 Scheduled command.

1. During a contact, the operator sends a command with a run time; the System answers "received".
2. At the run time, the command runs.
3. The "done" answer is stored and sent down at the next contact.

6.2 Off-Nominal Conditions

Grouped by NASA's kinds of off-nominal condition: low performance, operator errors, failures, and unexpected environmental conditions [1, App. S, p. 253]. Each entry gives the trigger, then the expected outcome.

Low performance of the radio link

Scenario-200 The radio link is lost during a firmware upload. The upload resumes at the next contact from where it stopped.

Scenario-210 Report storage fills during a long gap between contacts. At the next contact, the ground is told how many reports were lost.

Operator errors and unauthorized commands

Scenario-300 A command not from the operator, or an old command sent again, arrives. It is refused and logged [2, p. 62].

Scenario-310 A command not allowed in the current mode arrives, for example "switch" when no new firmware has arrived. It is refused with the reason [2, p. 63].

Scenario-320 A command with impossible values arrives. It is refused with the reason [2, p. 62].

Scenario-330 A dangerous command arrives without being armed. It is refused with the reason.

Scenario-340 A command arrives after its expiry. It is refused with the reason [2, p. 63].

Scenario-350 An answer is lost and the operator sends the command again. It is not run a second time; the answer is "already done".

Scenario-360 Firmware built for a different controller is uploaded. It is refused before it is installed [2, p. 63].

Firmware update failures

Scenario-400 Firmware is damaged in transfer. The onboard computer refuses it before it reaches the controller [2, p. 63].

Scenario-410 Firmware not released by the operator arrives. It is refused.

Scenario-420 Firmware older than the minimum version arrives. It is refused.

Scenario-430 The new firmware hangs during TRIAL. The controller returns to its previous firmware without outside help.

Scenario-440 The new firmware restarts over and over. After **[TBD-7]** failed starts, the controller returns to its previous firmware [2, p. 64].

Scenario-450 The new firmware runs but is never judged good. After the trial time, the controller returns to its previous firmware.

Scenario-460 The previous firmware is also unusable. The controller runs its last-resort firmware.

Scenario-470 The operator cancels an update in progress. The update stops; working firmware is left running.

Scenario-480 No firmware can run. The onboard computer restores the controller through the reprogramming link [2, p. 63].

Equipment failures

Scenario-500 One data-link connection fails. Everything continues on the other, and the failure is reported.

Scenario-510 Both data-link connections fail. The controller keeps working on its current firmware, the loss is reported, and the reprogramming link is still usable **[TBR-8]**.

Scenario-520 The controller goes silent. The onboard computer notices, tries to bring it back, and records what it did **[TBR-5]**.

Scenario-530 A program on the onboard computer crashes. It restarts and resumes any update in progress; the controller is not affected **[TBR-5]**.

Unexpected environmental conditions (from Section 4.0)

Scenario-600 Power is lost while the controller's firmware is being written. When power returns, the controller runs its previous firmware, and the update can be repeated [2, p. 63].

Scenario-610 A radiation bit error changes stored firmware. The controller detects it before starting that firmware and does not run it **[TBR-9]** [2, p. 63].

Scenario-620 A radiation bit error hits the memory the controller is running from. The controller may crash; it restarts on its own and resumes operation [2, p. 63].

7.0 Impact Considerations

Not applicable: bench demonstration.

8.0 Risks and Potential Issues

R-1 Both data-link connections may share software; one defect could cut both.

R-2 Controller memory may not hold last-resort plus two working firmwares.

R-3 Second data-link connection and reprogramming link not yet shown on bench.

R-4 Demonstration date **[TBD-2]**; undemonstrated scenarios are declared, not dropped.

A Acronyms

ConOps Concept of Operations

MOC Mission operations center

TBD To Be Determined

TBR To Be Resolved

TC Telecommand

TM Telemetry

B Glossary

Confirmed Firmware judged good after its trial; only confirmed firmware is known-good.

Last-resort Known-good firmware that updates never replace (golden image).

Switch The controller stops running old firmware and starts the new.

Arm First step of a dangerous command; changes nothing alone.

Expiry Time after which a waiting command is refused.

C Open Items

Unmarked statements were stated or accepted by the author on 9–10 October 2026. TBR items are resolved before the hazard analysis.

Resolved

TBR-1 Spacecraft controller, ground contacts only, no physical access, bench stand-in, bit errors occur, no repair. Accepted by the author, 10 October 2026.

TBR-2 Controller power can drop at any instant, including mid-write. Accepted by the author, 10 October 2026.

TBR-3 One operator. Accepted by the author, 10 October 2026.

TBR-4 Capability-1, Capability-2, Capability-3 and Capability-5 as written. Accepted by the author, 10 October 2026.

Open

TBR-5 O-4 and Capability-15 to Capability-18 as written (shown 10 Oct, not yet accepted).

TBR-6 Controller keeps working while new firmware arrives.
TBR-7 Onboard computer's own update out of scope.
TBR-8 Reprogramming link still usable with both data-link connections lost.
TBR-9 Stored firmware checked before start; damaged copy skipped.
TBR-10 Test setup can cut power and break one connection.
TBR-11 The engineer stays available to build and release new controller firmware for the life of the mission.
TBD-1 Project name and document reference.
TBD-2 Demonstration date.
TBD-3 Switch automatically on arrival, or only on operator command.
TBD-4 Who judges new firmware good: controller or operator.
TBD-5 Contact window length and gap (sourced, or a test parameter).
TBD-6 Trial time limit (requirements).
TBD-7 Failed starts before going back (requirements).
TBD-8 Update rate.
TBD-9 Temperature range.